Instructional Deviation As a Way to Drive Purposeful Learning

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INTRODUCTION

Common solutions to writer's block include changing one's environment, modifying a writing routine, or even simply taking a break. When a wall seems to be hit, this common block or lull in productivity is often solved by distinct changes in surroundings and applications that evoke new inspiration and ideas. These changes, which can be categorized as deviations to self and surroundings, allow individuals to make new discoveries and expand all-encompassing capabilities. This principle is not exclusive to individual practice, but also one that can be utilized in a classroom setting. Traditional teaching is typically lecture-based and teacher-centered, fatiguing students and dulling excitement towards new curriculum. By intentionally deviating from traditional teaching methods, students gain a purposeful learning experience that stimulates identity, pride, and bravery. These characteristics are important for students to feel their place in the classroom and walk away with meaningful education and experiences. This deviation can take on many forms: a change in teachers, a change in the environment where the lesson is being taught, and/or a change in the mode of teaching (kinesthetic learning approach). Whether these changes are executed independently or paired with another, they provide an opportunity to pursue their curiosity in a way that supports diverse applications of their existing knowledge that then contributes to the new lessons they would be learning.

RESEARCH QUESTIONS

How does educational disruption impact the students' learning outcomes? What is disruption, and how does it deviate from traditional learning? How do students' responses differ based on what disruption is implemented?

CONTEXT

SESSION A: MICROPLASTIC UNIT IN MARION JR/SR HIGH SCHOOL

Four Sustainability Fellows and three advisors focused on educational learning and microplastic research conducted a 4-week unit on microplastic research. The pilot unit was a first of its kind with its contents and findings to be replicated with other schools.

SESSION B: PLANET PROTECTOR CAMP IN SODUS JR/SR HIGH SCHOOL

5-day summer camp for middle schoolers mentored by graduate students and the Sustainability Fellows. The camp of focus for this research project was centered around invasive species found locally.

EDUCATIONAL METHODS



ARTIFACTS FROM ANALYSIS

Photographs, post it boards, posters, data tables, and graphic visualizations that showcased students' progress throughout their investigation were recorded.

"GOOD-OF-THE-ORDER" DEBRIEFS

Collaborative reflection was led by the education-focused advisors. The advisors collected Fellows' progress and ideation throughout the unit.

EVIDENCE RECORDINGS

Advisors and Fellows shared highlights of each day. The two education-bearing background advisors recorded dialogue for a comprehensive portfolio for future reference and research.

DISRUPTIONS AND THEIR CONSEQUENCES

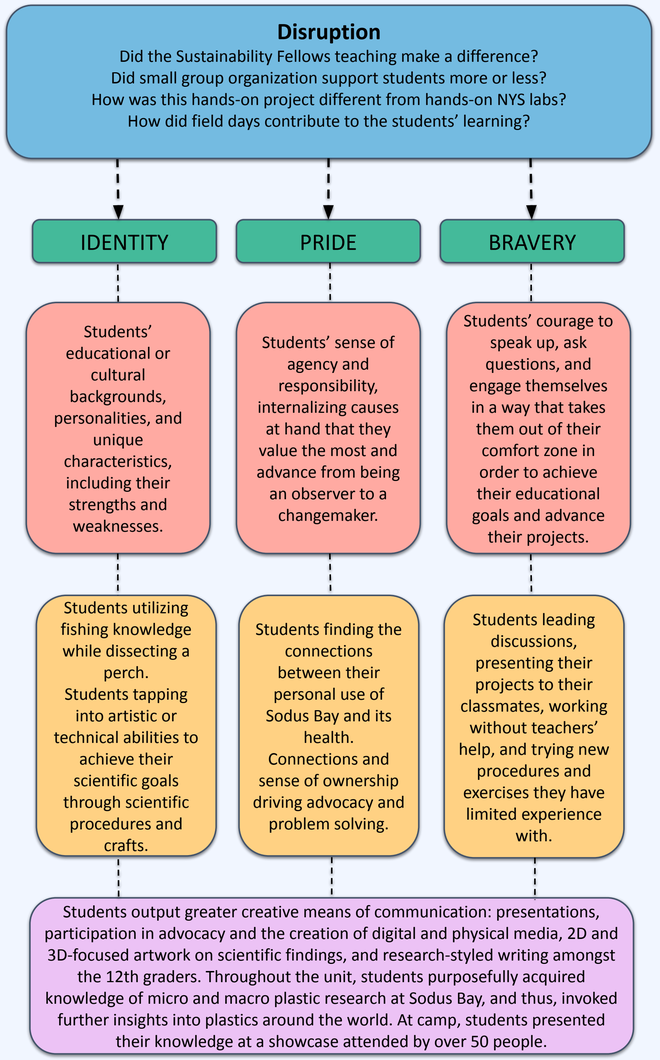




Image 1: Sustainability Fellows and Marion 8th graders holding PSA boards.



Image 2: Team *Invasive Investigators* creating "Nature Journals."



Image 3: Student presenting a demo of a boat collecting invasive species.

DISCUSSION

- Our five chosen methods of disruption resulted in purposeful learning which we believed was made evident by students expressing their identity, bravery, and pride.
  - **Location-Based Education:**
    - Identity: By focusing the scientific investigation around the local environment, Marion and Sodus students alike were able to pull from their existing knowledge of the area. By working in these students' backyard, a connection was able to be established and fostered throughout the project that personalized science.
    - Pride: With samples for the microplastic investigation coming directly from Sodus Bay and Lake Ontario, students were able to recognize the importance of the work they were completing as their findings had a direct impact on their personal lives. Students recognized that the places they were finding microplastics or invasive species were damaging places they, for example, swam or fished in.
  - **Fellows as Mentors**
    - Identity: Students and Fellows were able to connect on the "unknown": this was simultaneously a new *learning* experience for the students and a new *teaching* experience for the Fellows. Because the Fellows came from different backgrounds than the students, their identity was highlighted and utilized to personalize their projects.
  - **Kinesthetic Learning Approach:**
    - Pride: Hands-on activities gave students a sense of ownership over the things they were doing, evoking more care and attention. During an invasive species simulation, students built their own boats. When they saw invasive species were attaching to *their* boats, students swiftly began to think of solutions, consequently supporting the lesson plan on the mitigation of invasive species.
    - Bravery: Doing things with our hands requires courage to try new things, as opposed to listening to or watching someone else perform a procedure. At Marion, students had never dissected a fish before, and braving through the graphics and smells coming from the fish, students learned essential lessons in both dissections and fish anatomy.
  - **Student-Led Groups**
    - Bravery: Given complete freedom to conduct a research project is intimidating, especially if this is the first time ever having this kind of responsibility. Marion and Sodus students every day became more active in asking questions, proposing ideas, and participating in discussions about their projects and what they were learning.
    - Identity: With students leading their own projects and groups, their personalities, preferences, and backgrounds were the driving forces behind their projects and the goals of them.
  - **Small Groups, Close Relationships**
    - Bravery: Small groups of 2-5 students demonstrated more initiative to present their findings, lead investigations, and even loudly chant – "I have a recognition y'all" – to the whole science camp. By becoming closer to their peers and mentors, students felt more empowered to share their answers and ideas as there was an absence of judgement and a presence of encouragement.

CONCLUSIONS

While the particular focuses of the content Marion students did compared to Sodus student were different, — with one classroom being microplastics and the latter being invasive species — various educational disruptions were implemented. It was found that similar responses to content and material were being evoked, with students becoming increasingly active and engaged with each day. As students became more comfortable with the disruptions, the students inquired about the lesson plan contents, the ability to add their own disruptions to an activity, and even became comfortable to share previous material to a student who was absent — a tell-tale sign of true, purposeful learning. Thus, educational disruptions are imperative to providing students with meaningful experiences as it attaches itself to quality retention and critical thinking. Deviations from traditional teaching methods evoked various characteristics in students that led to an increase in participation and and engagement. Importantly, every deviation brought about a different set of characteristics. In order to maximize student learning when participating in non-traditional teaching styles, disruptions must be coupled, not isolated. This means that student-led groups should not be the only disruption that is enacted in the classroom, but rather, for example, student-led groups *and* location-based education. This helps set forth a framework for students to grasp: when students are placed in a dynamic learning environment, they come to understand that education does not have to be endured, but instead be a space that reflects, supports, and fosters their most authentic self, aiding in critical thinking and self-awareness in the classroom.

FURTHER WONDERINGS

This case study paves the way for instructors to include a deviation from their usual unit-teaching style. Teachers that can expect a full academic year of non-varied, traditional classwork, can insert disruptive modules to propagate student interests throughout the year. Further studies can implement a recurring disruptive unit and measure the demonstration of identity, pride, and bravery amongst students. Additional studies should be conducted on implementing disruptions in varying subjects like English, History, Math, or Art.

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